TrafficFlowBench 2026
Traffic State Reconstruction

STAEformer, ImputeFormer and an ensemble

Model comparison
Experiments 003–006

3 new Kaggle submissions
21 October 2026

Data and reconstruction pipeline

Synthetic highway observations · Task 1 reconstructs missing speed and flow.

Dataset

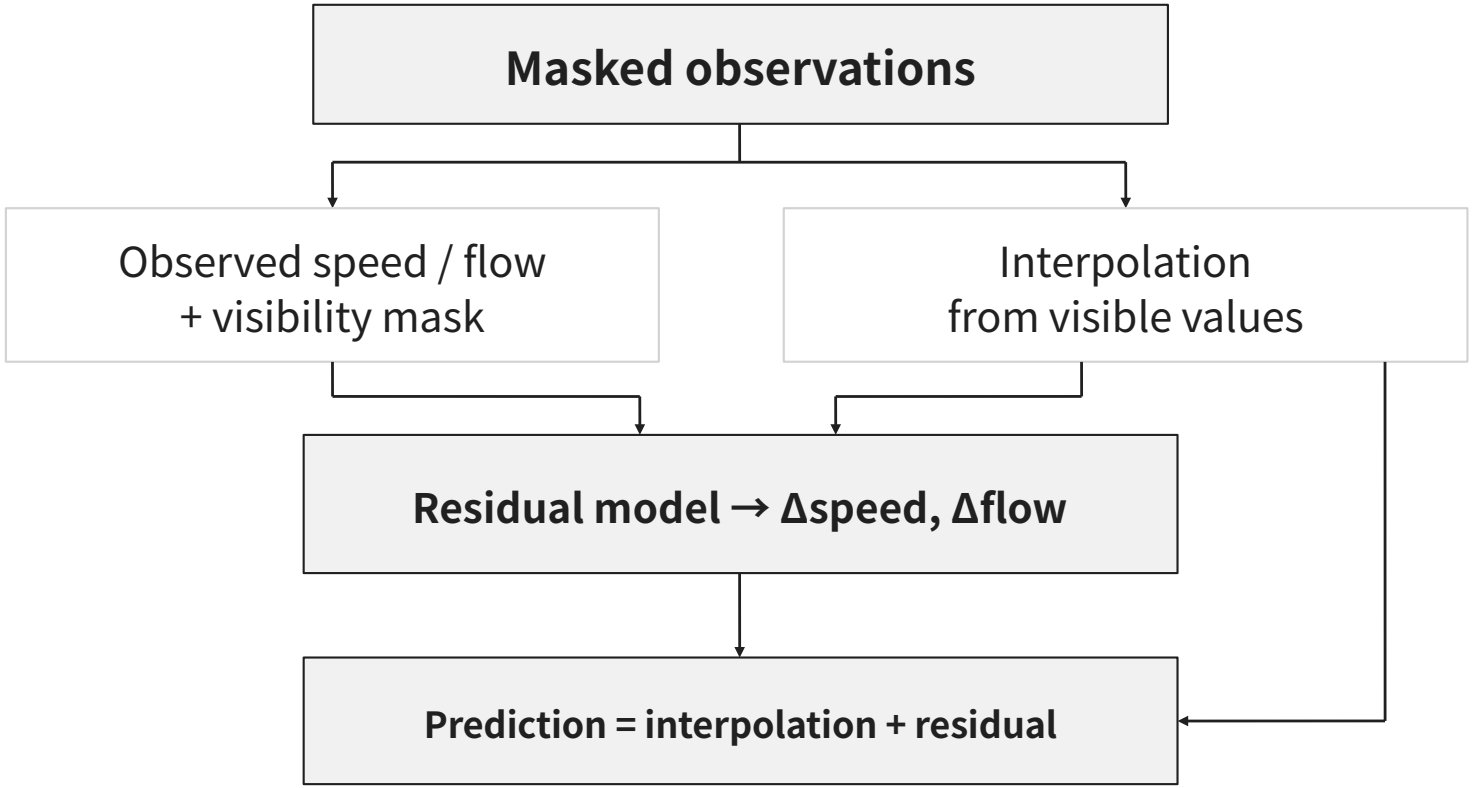
10 directional corridors

1,583 mainline links

5 min sampling interval

One day: 288 × N × C

N = links in one corridor
C = input channels; flow is per lane



Train with 20%, 30% or 50% random masking; compute loss only on hidden eligible cells.

Evaluate only the official Task 1 target coordinates. A blank cell is not automatically a target.

Models and evaluation setup

Two residual models, one common reconstruction pipeline.

Setting	Previous STAEformer	Improved STAEformer	ImputeFormer
Input window	24 steps / 2 h	48 steps / 4 h	24 steps / 2 h
Data channels	5	8	5
Training	5 epochs	5 + 15 epochs	20 epochs (from scratch)
Key difference	Reference model	Quality + time-gap inputs	Low-rank attention

New models: 50% overlapping windows · residual weight 1.0 · seed 42 · physical GPU 9

LOCAL FIT		SELECT MODEL / BLEND		REPORT RESULTS
Jun–Dec 2030	→	**Jan 2031**	→	**Feb 2031**

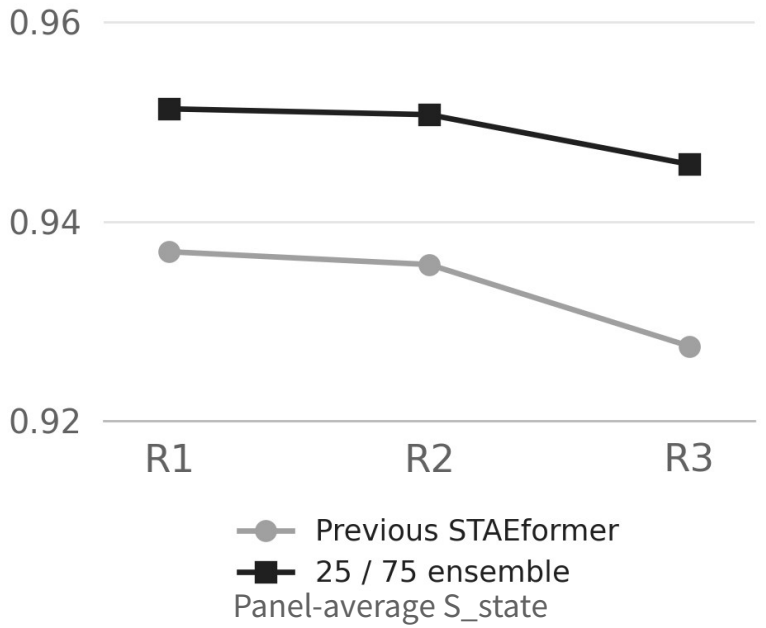
Final models: refit on all nine training months → predict the March / April inputs.

Local Task 1 results

S_state · higher is better · models trained on Jun–Dec 2030

Model	January selection	February report
Previous STAEformer	0.931830	0.933377
Improved STAEformer	0.943123	0.944272
ImputeFormer	0.947440	0.948697
25 / 75 ensemble	**0.948117**	**0.949252**

February by mask regime



25% STAEformer + 75% ImputeFormer

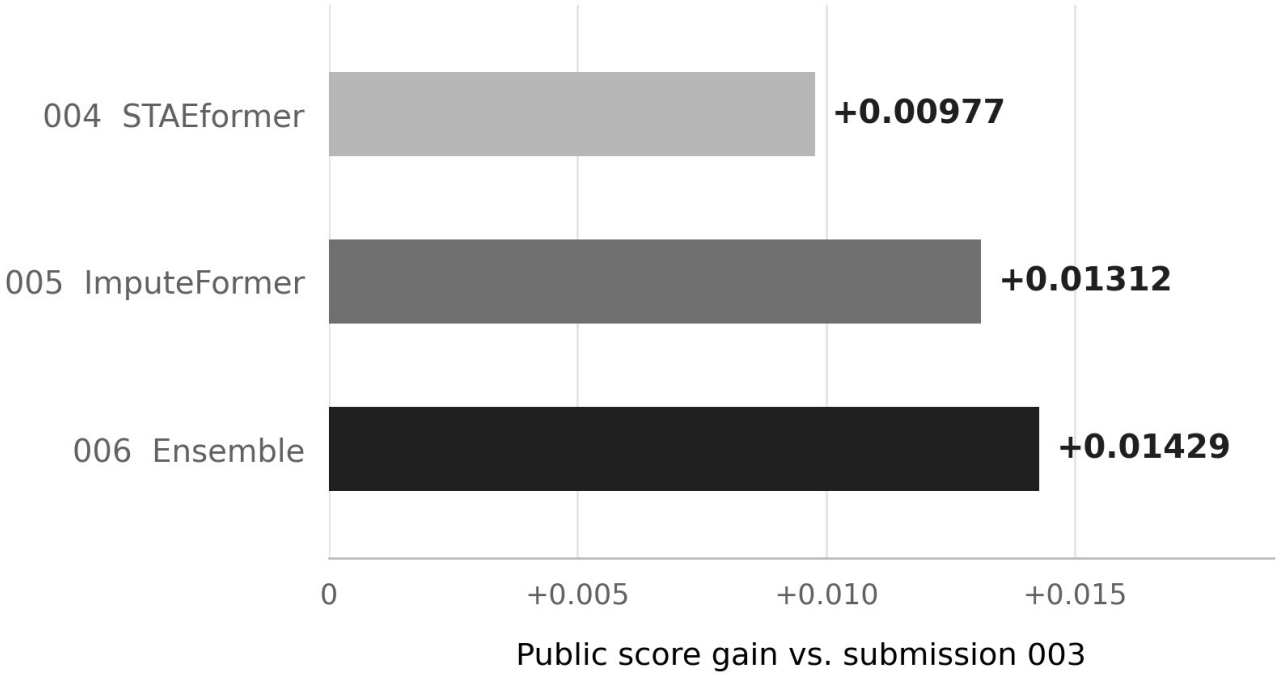
Blend selected on January · all 3,443,944 February targets evaluated

February was also evaluated in earlier experiments; this experiment used January for selection.

Kaggle Public results

Three new submissions completed · Task 2 (queue) and Task 4 (path flow) held fixed

Improvement over submission 003



Submission	Public score
003 Previous	0.76826
004 STAEformer	0.77803
005 ImputeFormer	0.78138
006 Ensemble	**0.78255**

+0.01429

Best gain over the previous submission

The Public gain reflects Task 1 reconstruction + Task 3 physical consistency.

Conclusion and next experiment

25% STAEformer + 75% ImputeFormer

Best of the four configurations tested.

0.78255

Best Kaggle Public score

+0.01429 over submission 003

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01 ImputeFormer was the strongest single model.

02 The ensemble added +0.00117 on the Public leaderboard.

03 **Next: train with time-block and spatial masks.**

Then test window length, extra features and training duration one at a time.

Private leaderboard results are not yet available.

IMAI